Strange Attractors

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Strange Attractors

- Definition
- Application to this project
- Significance

Lorenz Attractor

- Background (to be done later)
- The Lorenz equations originate from Rayleigh–Bénard convection,. The model assumes the Boussinesq approximation.
- The velocity and temperature fields are expanded in spatial Fourier modes. Retaining only three dominant modes reduces the original partial differential equations to a finite-dimensional dynamical system.
- The quadratic nonlinearities arise from the nonlinear advection terms.

$$\dot{X} = \sigma(Y - X), \dot{Y} = rX - Y - XZ, \dot{Z} = XY - bZ.$$

- X: convection amplitude
- Y: horizontal temperature variation
- Z: distortion of the vertical temperature profile

Thomas Attractor

- Background (to be done later)
- The Thomas attractor is a simple three-dimensional model of cyclically symmetric nonlinear dynamics. It is primarily a phenomenological dynamical system.

$$\dot{x} = \sin y - bx, \dot{y} = \sin z - by, \dot{z} = \sin x - bz.$$

- The divergence of the vector field is $\nabla \cdot \mathbf{F} = -3b$.
- For b>0, the phase-space volume contracts exponentially. The system may exhibit different dynamical behaviors, including chaotic motion.
- The Thomas system combines cyclic nonlinear feedback, represented by the sine terms, with linear dissipation. Their competition can produce bounded, complex trajectories and, for suitable parameter values, a strange attractor.

Langford Attractor

- Background (to be done later)
- Langford equations are constructed to investigate the interplay between nonlinear oscillations, symmetry, and dissipation.
- One may use this form

$$\dot{x} = (z - \beta)x - \omega y,$$
$$\dot{y} = \omega x + (z - \beta)y,$$
$$\dot{z} = \lambda + \alpha z - \frac{z^3}{3} - (x^2 + y^2)(1 + \gamma z) + \delta z x^3.$$

- The first two equations describe coupled oscillatory dynamics in the x-y plane, with z modulating the growth or decay of the oscillations. The third equation governs the evolution of z through nonlinear feedback and saturation.
- The existence of a bounded attractor depends on the parameter values and the balance between nonlinear feedback and dissipation.

Rössler Attractor

- Background (to be done later)
- The Rössler attractor is a minimal three-dimensional dynamical system capable of producing chaos.

$$\dot{x} = -y - z, \dot{y} = x + ay, \dot{z} = b + z(x - c).$$

- The first two equations generate approximately rotational motion in the x-y plane. The third equation introduces nonlinear feedback through the term xz, which can drive the trajectory away from the plane and produce a sudden excursion before it returns.
- The divergence of the vector field is $\nabla \cdot \mathbf{F} = a + x - c.$
- The long-term average divergence along typical trajectories may be negative, indicating average phase-space volume contraction. The competition between rotational motion, nonlinear feedback, and dissipation can generate a strange attractor.
- It typically exhibits a spiral structure with a single dominant lobe. Trajectories wind around an unstable region repeatedly, while occasional excursions in the third dimension.